# HUMAN 2.0

## THE EVOLUTION OF OUR SPECIES MIGHT, FOR THE VERY FIRST TIME, BE DONE ON PURPOSE

BY ROBERT SOVEREIGN-SMITH

It's one thing to genetically engineer crops, or sheep, or earthworms... but doing it to humans opens up a whole new floodgate. There's really no black or white, clear right and clear wrong in such cases, which is why this is such dangerous territory that we're treading.

It's like wading through a nuclear minefield, and every step we take gives us added confidence of our safety and righteousness, but we're still just one step away from annihilating the world as we know it.

Before we delve into this month's cover story, however, we should tell you that this isn't going to be a medical scientist's take, or be filled with medical jargon – mostly because we're not qualified to talk about complex genetic engineering, and don't even fully understand how it's done... but also because "What can we do?" is not the important question, whereas "What should we do?" is.

### CHINESE EMBRYOS

The genetics world was shocked when Chinese scientists at Sun Yat-sen University in Guangzhou, announced that they had successfully engineered human embryos to modify a gene that we think is responsible for thalassaemia – a fatal disease that causes improper production of haemoglobin, and eventually death.

The scientists claimed that they had not taken viable embryos, and thus, this was not a violation of globally accepted ethics regarding cloning, genetics and the like. They used embryos sourced from IVF clinics that would otherwise be destroyed because they were "non-viable" – these were eggs that had been fertilised by 2 sperm, and thus could never result in a baby anyway.

Much like you do in a word processor, they were able to cut and paste genetic code – using a technique called CRISPR/Cas9 to remove the bad genetic code and add in new code. The results were not always good, because sometimes the wrong codes were getting cut out and replaced, which basically means a very genetically deformed embryo, but that's just a question of honing their skills, because they also did get it right.

Let's be clear, they are not claiming that they have cured thalassemia; they're saying they could, in theory, make sure that if two people want to reproduce, they can make sure their baby will not have thalassemia. Of course, this can only be perfected by allowing embryos to grow and form humans in order to investigate the results, and thankfully the world has outlawed that as of now.

### CRISPR?

Clustered regularly-interspaced short palindromic repeats (CRISPR) is a defense system used by bacteria to protect themselves against viruses. Viruses infect cells by injecting their DNA into them, and bacteria protect themselves by producing short RNA strands, which have a sequence that matches the virus DNA in question. Cas9 is a protein nucleus that is capable of breaking down and editing DNA. The bacteria sends out the RNA strands with, amongst others, the Cas9 protein (Cas = CRISPR-associated). This then finds the viral dna, bonds the bacteria-created RNA to the viral dna, and this causes it to mutate, or die out because it isn't able to replicate. Controlling this method is what is used to edit dna of everything from simple viruses to humans.

### DESIGNER BABIES

If genetic editing is the spark, then our human flaws are the fuel. It's one thing to find a cure for a disease, and totally another to try and engineer flaws out of people. What will start off as trying to cure diseases will end up with people wanting to design their offspring. It's a natural and logical progression, isn't it? It will start with rich people wanting to make sure that they have healthy kids, and then move on to someone willing to pay even more to ensure that they get a child with the muscular attributes of, say, Usain Bolt. Some might want a kid that looks like, say, Alia Bhatt, but has the intellect of Einstein – wouldn't that be weird?

India is a prime example of how cultural selection itself leads people to